

The Sherman–Morrison–Woodbury Formula and Its Applications in Data Science

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1 Introduction

Solving a linear system of the form

$$Ax = b$$

is the ultimate end point of many procedures in numerical linear algebra, scientific computing, statistics, and data science. In many applications, a complicated computational procedure eventually reduces to the problem of solving such a linear system.

If the matrix A is nonsingular, then theoretically the solution may be written as

$$x = A^{-1}b.$$

In practical computation, of course, one generally does not explicitly compute A^{-1} merely for the purpose of solving a linear system. Nevertheless, the inverse notation is extremely useful for understanding how a solution changes when the underlying problem changes.

The vector b may change from one problem to another. However, in many important applications, the matrix A itself may also change. This is particularly common in data science, where the matrices under consideration are constructed from data. New observations may be added, old observations may be removed, weights may be changed, or a model may be updated as new information becomes available.

Suppose that we have already performed the expensive computation associated with a matrix A , and then A is modified to a new matrix

$$\tilde{A} = A + \Delta A.$$

One possibility is simply to forget everything we previously computed and solve the new problem from the beginning. However, this may be extremely wasteful, particularly when A is very large and the change ΔA possesses some simple structure.

This naturally leads to the following question:

If a matrix changes in a structured way, can we use what we already know about the original matrix to efficiently obtain information about the modified matrix?

The Sherman–Morrison and Woodbury formulas provide a remarkably elegant answer to this question.

2 Rank-One Changes

The simplest structured modification of a matrix is a rank-one change.

Let

$$u, v \in \mathbb{R}^n.$$

Then the outer product

$$uv^T$$

is an $n \times n$ matrix. Provided that u and v are nonzero, this matrix has rank one.

Therefore, if the change in A can be represented as

$$\Delta A = uv^T,$$

then the modified matrix becomes

$$\tilde{A} = A + uv^T.$$

Similarly, we may encounter

$$\tilde{A} = A - uv^T.$$

An important point should be emphasized here. A *rank-one change* does not necessarily mean that only one entry of the matrix has changed, nor does it mean that the numerical change must be small. In fact, uv^T may change every entry of A . What matters is the structure of the change, namely,

$$\text{rank}(\Delta A) = 1.$$

For example,

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

Every entry of the resulting matrix is nonzero, but the matrix still has rank one.

Thus, the number of entries that change is not the important issue. The important issue is the *rank of the change*.

3 From Rank-One Changes to General Low-Rank Changes

A change in A may, of course, have rank greater than one. Suppose

$$\text{rank}(\Delta A) = k.$$

Such a change may be expressed as a sum of rank-one matrices,

$$\Delta A = u_1 v_1^T + u_2 v_2^T + \cdots + u_k v_k^T.$$

Consequently,

$$\tilde{A} = A + u_1 v_1^T + u_2 v_2^T + \cdots + u_k v_k^T.$$

At this point there are two natural possibilities.

First, one may update the matrix one rank-one modification at a time. This leads naturally to the **Sherman–Morrison formula**.

Second, we may collect all of the rank-one modifications together. Define

$$U = \begin{bmatrix} u_1 & u_2 & \cdots & u_k \end{bmatrix}, \quad V = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}.$$

Then

$$UV^T = u_1v_1^T + u_2v_2^T + \cdots + u_kv_k^T,$$

and therefore

$$\boxed{\tilde{A} = A + UV^T.}$$

This leads to the more general **Woodbury matrix identity**.

The basic philosophy behind both formulas can therefore be summarized as

$$\boxed{\text{Do not recompute everything when only a structured part has changed.}}$$

The Sherman–Morrison formula handles a rank-one modification, while the Woodbury formula extends the same idea to a low-rank modification.

In the following sections, we shall derive these formulas directly from the linear system

$$Ax = b,$$

rather than simply presenting them as identities to be memorized. This approach makes clear both where the formulas come from and why they are computationally useful.

4 The Sherman–Morrison Formula

We now consider the simplest possible low-rank modification of a matrix: a rank-one update. Suppose that $A \in \mathbb{R}^{n \times n}$ is nonsingular and that A^{-1} is already known. Let $u, v \in \mathbb{R}^n$, and suppose that A is modified according to

$$\tilde{A} = A - uv^T.$$

Our objective is to determine the inverse of the modified matrix,

$$(A - uv^T)^{-1},$$

without computing this inverse again from the beginning.

Rather than attempting to guess the formula for the new inverse, we shall derive it directly by asking a much more natural question:

How does the solution of $Ax = b$ change when A is replaced by $A - uv^T$?

4.1 Beginning with the Modified Linear System

Consider

$$(A - uv^T)x = b.$$

Expanding the left-hand side gives

$$Ax - uv^Tx = b.$$

Since A^{-1} is assumed to be known, premultiply the equation by A^{-1} :

$$A^{-1}Ax - A^{-1}uv^Tx = A^{-1}b.$$

Using

$$A^{-1}A = I,$$

we obtain

$$x - A^{-1}uv^T x = A^{-1}b.$$

To simplify the notation, define

$$z = A^{-1}u$$

and

$$y = A^{-1}b.$$

Therefore,

$$\boxed{x - zv^T x = y.}$$

4.2 The Important Observation: $v^T x$ is a Scalar

At first sight, the equation

$$x - zv^T x = y$$

still contains the unknown vector x in two places. However, the second appearance of x has a very special form.

Since

$$v^T \in \mathbb{R}^{1 \times n}$$

and

$$x \in \mathbb{R}^{n \times 1},$$

we have

$$v^T x \in \mathbb{R}.$$

Thus, $v^T x$ is not another unknown vector. It is merely a scalar.

Let

$$\boxed{\alpha = v^T x.}$$

Then

$$x - z\alpha = y,$$

and consequently

$$\boxed{x = y + \alpha z.}$$

This equation reveals the basic geometry of the rank-one update. The vector

$$y = A^{-1}b$$

is the solution associated with the original matrix A , while the new solution differs from y only by a scalar multiple of

$$z = A^{-1}u.$$

In other words,

$$\boxed{\text{new solution} = \text{old solution} + \text{scalar correction} \times \text{correction direction}.}$$

The entire effect of the rank-one modification has therefore been reduced to determining a single scalar α .

4.3 Solving for the Scalar α

We know that

$$x - z\alpha = y.$$

Premultiply this equation by v^T :

$$v^T x - v^T z \alpha = v^T y.$$

But by definition,

$$v^T x = \alpha.$$

Therefore,

$$\alpha - v^T z \alpha = v^T y.$$

Since $v^T z$ is also a scalar, we may factor α :

$$(1 - v^T z)\alpha = v^T y.$$

Provided that

$$1 - v^T z \neq 0,$$

we obtain

$$\boxed{\alpha = \frac{v^T y}{1 - v^T z}.$$

Now substitute

$$z = A^{-1}u$$

and

$$y = A^{-1}b.$$

Then

$$\alpha = \frac{v^T A^{-1}b}{1 - v^T A^{-1}u}.$$

4.4 Recovering the New Solution

Recall that

$$x = y + \alpha z.$$

Substituting the expressions for y , z , and α gives

$$x = A^{-1}b + A^{-1}u \frac{v^T A^{-1}b}{1 - v^T A^{-1}u}.$$

Equivalently,

$$x = A^{-1}b + A^{-1}u (1 - v^T A^{-1}u)^{-1} v^T A^{-1}b.$$

Now factor b from the right:

$$x = \left[A^{-1} + A^{-1}u (1 - v^T A^{-1}u)^{-1} v^T A^{-1} \right] b.$$

On the other hand, the modified system was

$$(A - uv^T)x = b,$$

so

$$x = (A - uv^T)^{-1}b.$$

Since the preceding relation holds for an arbitrary vector b , we obtain

$$(A - uv^T)^{-1} = A^{-1} + A^{-1}u(1 - v^T A^{-1}u)^{-1}v^T A^{-1}.$$

This is the **Sherman–Morrison formula** for a rank-one subtractive update.

4.5 The More Common Additive Form

The formula is frequently written for the update

$$\tilde{A} = A + uv^T.$$

Replacing u by $-u$ in the preceding result gives

$$(A + uv^T)^{-1} = A^{-1} - A^{-1}u(1 + v^T A^{-1}u)^{-1}v^T A^{-1}.$$

Since

$$1 + v^T A^{-1}u$$

is a scalar, this may also be written as

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}.$$

4.6 When Does the Formula Fail?

The Sherman–Morrison formula requires

$$1 + v^T A^{-1}u \neq 0$$

for the additive form, or

$$1 - v^T A^{-1}u \neq 0$$

for the subtractive form.

This condition is not merely an algebraic inconvenience. If

$$1 + v^T A^{-1}u = 0,$$

then the updated matrix

$$A + uv^T$$

is singular, and consequently its inverse does not exist.

Thus, the denominator appearing in the Sherman–Morrison formula also provides information about whether the rank-one modification destroys the invertibility of the original matrix.

4.7 Interpretation of the Formula

The Sherman–Morrison formula may be viewed schematically as

$$\boxed{\text{new inverse} = \text{old inverse} - \text{rank-one correction.}}$$

Indeed,

$$A^{-1}u$$

is an $n \times 1$ vector and

$$v^T A^{-1}$$

is a $1 \times n$ vector. Therefore,

$$A^{-1}u v^T A^{-1}$$

is itself a rank-one matrix.

Hence, a rank-one modification of A produces a rank-one correction to A^{-1} , provided the modified matrix remains nonsingular.

This is the essential computational idea behind the formula. If the information associated with A has already been computed, then a structured change in A should, whenever possible, be handled by an efficient update rather than by discarding the previous computation and starting again.

5 A Numerical Example of the Sherman–Morrison Formula

We now illustrate the Sherman–Morrison formula using a simple 2×2 example.

Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

We consider the subtractive rank-one update

$$\tilde{A} = A - uv^T.$$

First compute

$$uv^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

Therefore,

$$\tilde{A} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix}.$$

Our objective is to compute

$$\tilde{A}^{-1} = (A - uv^T)^{-1}$$

using the Sherman–Morrison formula.

For the subtractive case,

$$(A - uv^T)^{-1} = A^{-1} + A^{-1}u(1 - v^T A^{-1}u)^{-1}v^T A^{-1}.$$

5.1 Step 1: Compute A^{-1}

Since A is diagonal,

$$A^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}.$$

5.2 Step 2: Compute $A^{-1}u$

We obtain

$$A^{-1}u = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix}.$$

5.3 Step 3: Compute $v^T A^{-1}$

Similarly,

$$v^T A^{-1} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \end{bmatrix}.$$

5.4 Step 4: Compute the Scalar Denominator

We next compute

$$v^T A^{-1}u.$$

Using the result from the previous step,

$$v^T A^{-1}u = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} = \frac{1}{2}.$$

Therefore,

$$1 - v^T A^{-1}u = 1 - \frac{1}{2} = \frac{1}{2},$$

and hence

$$(1 - v^T A^{-1}u)^{-1} = 2.$$

5.5 Step 5: Compute the Rank-One Correction

The correction term is

$$A^{-1}u (1 - v^T A^{-1}u)^{-1} v^T A^{-1}.$$

Substituting the quantities already computed gives

$$\begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} (2) \begin{bmatrix} 1/2 & 0 \end{bmatrix}.$$

Since

$$2 \begin{bmatrix} 1/2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

the correction becomes

$$\begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 1/3 & 0 \end{bmatrix}.$$

5.6 Step 6: Obtain the Updated Inverse

Therefore,

$$(A - uv^T)^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 1/3 & 0 \end{bmatrix}.$$

Thus,

$$\boxed{(A - uv^T)^{-1} = \begin{bmatrix} 1 & 0 \\ 1/3 & 1/3 \end{bmatrix}.$$

5.7 Direct Verification

We can verify the result directly by multiplication:

$$\begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Hence,

$$\boxed{\tilde{A}\tilde{A}^{-1} = I,}$$

confirming that the Sherman–Morrison formula gives the correct inverse.

5.8 What This Example Shows

The important point is not merely that the arithmetic works. The example illustrates the structure of the update:

$$\boxed{\text{new inverse} = \text{old inverse} + \text{rank-one correction}.}$$

We did not need to derive the inverse of the modified matrix independently. Instead, we used the inverse already known for A and corrected it using only the vectors u and v .

For a 2×2 matrix this may not appear impressive. However, when A is extremely large and the modification has low rank, this same idea can save a substantial amount of computation.

6 The Woodbury Matrix Identity

The Sherman–Morrison formula handles a rank-one update of the form

$$A + uv^T.$$

Suppose now that the change in A has rank greater than one. Let

$$U, V \in \mathbb{R}^{n \times k},$$

where typically

$$k \ll n.$$

Then a rank- k modification may be written in the form

$$\tilde{A} = A + UV^T.$$

The objective is again to determine the inverse of the modified matrix without recomputing everything from the beginning.

Rather than memorizing the Woodbury formula, we shall derive it directly from the modified linear system

$$(A + UV^T)x = b.$$

6.1 Beginning with the Modified System

Consider

$$(A + UV^T)x = b.$$

Expanding gives

$$Ax + UV^T x = b.$$

Assuming that A^{-1} is already known, premultiply by A^{-1} :

$$A^{-1}Ax + A^{-1}UV^T x = A^{-1}b.$$

Therefore,

$$x + A^{-1}UV^T x = A^{-1}b.$$

Define

$$Z = A^{-1}U$$

and

$$y = A^{-1}b.$$

Then

$$\boxed{x + ZV^T x = y.}$$

6.2 The Key Difference from Sherman–Morrison

In the Sherman–Morrison derivation, the quantity

$$v^T x$$

was a scalar.

Here,

$$V^T x$$

is generally not a scalar.

Since

$$V^T \in \mathbb{R}^{k \times n}$$

and

$$x \in \mathbb{R}^{n \times 1},$$

we have

$$V^T x \in \mathbb{R}^{k \times 1}.$$

Thus, instead of introducing one scalar unknown, we introduce a small vector.

Let

$$\boxed{\alpha = V^T x,}$$

where

$$\alpha \in \mathbb{R}^k.$$

Then

$$x + Z\alpha = y,$$

and consequently

$$\boxed{x = y - Z\alpha.}$$

The original unknown vector x may have dimension n , but the correction has now been reduced to determining only the k components of α .

6.3 Solving for the Correction Vector

Starting from

$$x + Z\alpha = y,$$

premultiply by V^T :

$$V^T x + V^T Z\alpha = V^T y.$$

Since

$$V^T x = \alpha,$$

we obtain

$$\alpha + V^T Z\alpha = V^T y.$$

Factor α :

$$(I + V^T Z)\alpha = V^T y.$$

Provided that

$$I + V^T Z$$

is nonsingular, we obtain

$$\alpha = (I + V^T Z)^{-1} V^T y.$$

Now substitute

$$Z = A^{-1}U$$

and

$$y = A^{-1}b.$$

Then

$$\alpha = (I + V^T A^{-1}U)^{-1} V^T A^{-1}b.$$

6.4 Recovering the New Solution

Recall that

$$x = y - Z\alpha.$$

Substituting the expressions above gives

$$x = A^{-1}b - A^{-1}U (I + V^T A^{-1}U)^{-1} V^T A^{-1}b.$$

Factor b from the right:

$$x = \left[A^{-1} - A^{-1}U (I + V^T A^{-1}U)^{-1} V^T A^{-1} \right] b.$$

On the other hand,

$$x = (A + UV^T)^{-1}b.$$

Therefore,

$$(A + UV^T)^{-1} = A^{-1} - A^{-1}U (I + V^T A^{-1}U)^{-1} V^T A^{-1}.$$

This is the **Woodbury matrix identity**.

6.5 Why the Formula is Computationally Attractive

Suppose

$$A \in \mathbb{R}^{n \times n},$$

while

$$U, V \in \mathbb{R}^{n \times k}.$$

Then

$$V^T A^{-1} U$$

has dimension

$$(k \times n)(n \times n)(n \times k) = k \times k.$$

Therefore,

$$I + V^T A^{-1} U$$

is only a $k \times k$ matrix.

This is the central computational advantage of the Woodbury formula.

If

$$n = 100000$$

but

$$k = 10,$$

then the matrix appearing in the middle inverse has size only

$$10 \times 10.$$

Thus, a modification to a very large matrix can sometimes be handled by solving a much smaller problem.

6.6 Relationship with Sherman–Morrison

The Sherman–Morrison formula is contained inside the Woodbury identity as the special case

$$k = 1.$$

If

$$U = u$$

and

$$V = v,$$

then

$$V^T A^{-1} U = v^T A^{-1} u$$

is a scalar.

Hence,

$$I + V^T A^{-1} U$$

reduces to

$$1 + v^T A^{-1} u.$$

Therefore the Woodbury formula becomes

$$(A + uv^T)^{-1} = A^{-1} - A^{-1}u(1 + v^T A^{-1}u)^{-1}v^T A^{-1},$$

which is precisely the Sherman–Morrison formula.

Thus,

$$\boxed{\text{Sherman–Morrison} = \text{rank-one Woodbury.}}$$

Equivalently,

$$\boxed{\text{Woodbury} = \text{Sherman–Morrison generalized to rank } k.}$$

6.7 The Main Idea

The Woodbury identity may be viewed schematically as

$$\boxed{\text{new inverse} = \text{old inverse} - \text{low-rank correction.}}$$

The essential lesson is therefore the same as before:

$$\boxed{\text{Exploit the structure of the change instead of recomputing everything.}}$$

When the modification has low rank relative to the size of the original matrix, this principle can lead to a substantial reduction in computational work.

7 Numerical Considerations and a Clever Computational Technique

The Sherman–Morrison and Woodbury identities are commonly written in terms of A^{-1} . From a theoretical point of view, this notation is natural and convenient. From a numerical point of view, however, an important question immediately arises:

Do we actually need to compute the inverse A^{-1} in order to use the Woodbury formula?

The answer is no.

This distinction is extremely important in numerical linear algebra and data science. If our objective is to evaluate

$$A^{-1}b,$$

we generally should not first compute the entire matrix A^{-1} and then multiply it by b . Instead, we solve

$$Ax = b.$$

Similarly, if several systems involving the same matrix A must be solved, we may factorize A once—for example using LU, QR, or another appropriate factorization—and then reuse that factorization for different right-hand sides.

Thus, although the Woodbury identity contains several appearances of A^{-1} , its practical implementation need not explicitly construct A^{-1} at all.

This leads to a particularly clever computational interpretation of the formula.

7.1 The General Woodbury Formula

The Woodbury identity is named after Max A. Woodbury, who studied the problem of inverting modified matrices.

Consider the more general modification

$$\tilde{A} = A + BCD,$$

where

$$\begin{aligned} A &\in \mathbb{R}^{n \times n}, & B &\in \mathbb{R}^{n \times m}, \\ C &\in \mathbb{R}^{m \times m}, & D &\in \mathbb{R}^{m \times n}. \end{aligned}$$

Assume that the required matrices are nonsingular. Then

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}.$$

This is the general form of the Woodbury matrix identity.

The version derived previously,

$$(A + UV^T)^{-1} = A^{-1} - A^{-1}U(I + V^T A^{-1}U)^{-1}V^T A^{-1},$$

is obtained immediately by choosing

$$B = U, \quad C = I, \quad D = V^T.$$

Indeed,

$$BCD = UIV^T = UV^T.$$

Thus the formula involving $A + UV^T$ is simply a particularly convenient special case of the more general identity.

7.2 The Numerical Problem

Suppose our actual objective is not to construct

$$(A + BCD)^{-1}$$

but rather to solve

$$(A + BCD)x = b.$$

One possibility would be to compute the Woodbury formula literally. However, that expression contains

$$A^{-1},$$

and explicitly constructing a large inverse may itself be expensive and unnecessary.

Instead, we shall convert every occurrence of A^{-1} into the solution of a linear system.

This produces a practical algorithm that uses the Woodbury idea without ever explicitly forming A^{-1} .

7.3 Step 1: Solve the Original System

First solve

$$\boxed{Ad = b}$$

for d .

Theoretically,

$$d = A^{-1}b.$$

Numerically, however, we do not need to compute A^{-1} . We simply solve the system $Ad = b$.

Thus d represents the solution associated with the original matrix A before the modification BCD is introduced.

7.4 Step 2: Compute $E = A^{-1}B$ Without Computing A^{-1}

Next we require the quantity

$$A^{-1}B.$$

Define

$$\boxed{E = A^{-1}B.}$$

Suppose the columns of B are

$$B = [B_1 \ B_2 \ \cdots \ B_m].$$

Then

$$E = [A^{-1}B_1 \ A^{-1}B_2 \ \cdots \ A^{-1}B_m].$$

Therefore, instead of computing A^{-1} , solve

$$Az_i = B_i, \quad i = 1, \dots, m.$$

The resulting vectors are collected as

$$E = [z_1 \ z_2 \ \cdots \ z_m].$$

Equivalently, this may be viewed as solving the matrix equation

$$\boxed{AE = B.}$$

Thus,

$$E = A^{-1}B$$

has been obtained entirely through linear solves.

7.5 Step 3: Introduce a Small Auxiliary Variable

Return to the modified system

$$(A + BCD)x = b.$$

Expanding gives

$$Ax + BCDx = b.$$

Move the correction term to the right:

$$Ax = b - BCDx.$$

Define

$$\boxed{y = CDx.}$$

Then

$$BCDx = By,$$

and consequently

$$\boxed{Ax = b - By.}$$

If y were already known, this would immediately give the final solution x by solving one more system involving A .

Thus, the remaining problem is to determine the much smaller vector

$$y \in \mathbb{R}^m.$$

7.6 Step 4: Derive the Small $m \times m$ System

Since

$$y = CDx,$$

and C is assumed nonsingular,

$$C^{-1}y = Dx.$$

From

$$Ax = b - By,$$

we may write theoretically

$$x = A^{-1}(b - By).$$

Therefore,

$$x = A^{-1}b - A^{-1}By.$$

But we have already computed

$$d = A^{-1}b$$

and

$$E = A^{-1}B.$$

Hence,

$$\boxed{x = d - Ey.}$$

Multiply by D :

$$Dx = Dd - DEy.$$

But

$$Dx = C^{-1}y.$$

Therefore,

$$C^{-1}y = Dd - DEy.$$

Move the final term to the left:

$$C^{-1}y + DEy = Dd.$$

Factor y :

$$\boxed{(C^{-1} + DE)y = Dd.}$$

This is the crucial reduction.

The original problem involved an $n \times n$ matrix, but the new system

$$(C^{-1} + DE)y = Dd$$

has dimension only

$$\boxed{m \times m.}$$

When

$$m \ll n,$$

this can represent a very substantial computational saving.

7.7 Step 5: Recover the Final Solution

After solving

$$(C^{-1} + DE)y = Dd$$

for y , return to

$$Ax = b - By.$$

Then solve

$$\boxed{Ax = b - By}$$

for the final solution x .

Again, no explicit inverse of A is required.

7.8 The Complete Computational Procedure

The entire procedure may therefore be summarized as follows:

$$\boxed{\begin{aligned} Ad &= b, \\ AE &= B, \\ (C^{-1} + DE)y &= Dd, \\ Ax &= b - By. \end{aligned}}$$

The corresponding theoretical quantities are

$$d = A^{-1}b, \quad E = A^{-1}B,$$

but neither requires the explicit construction of A^{-1} .

If a factorization of A is already available, the same factorization can be reused for the systems

$$Ad = b, \quad AE = B, \quad Ax = b - By.$$

This reuse is one of the principal numerical advantages of the method.

7.9 Why This Technique is Clever

At first sight, the Woodbury identity appears to require knowledge of the inverse

$$A^{-1}.$$

A literal reading of the formula might therefore suggest that we must first construct this inverse. However, the numerical interpretation is more subtle.

Whenever we see an expression such as

$$A^{-1}r,$$

we should ask whether it can instead be interpreted as the solution of

$$Az = r.$$

Thus,

$$\boxed{A^{-1}r \longleftrightarrow \text{solve } Az = r.}$$

This simple change in viewpoint transforms an identity involving matrix inverses into a practical algorithm involving linear system solves.

Furthermore, the new information introduced by the modification is handled through the small system

$$(C^{-1} + DE)y = Dd,$$

whose dimension is only m .

Therefore the real computational structure is

$$\boxed{\begin{array}{c} \text{reuse the original large system} \\ + \\ \text{solve one small correction problem} \\ + \\ \text{recover the updated solution.} \end{array}}$$

This is a beautiful example of the distinction between a mathematical formula and a numerical algorithm. The formula tells us what the answer is; numerical reasoning tells us how to obtain that answer without performing unnecessary computation.

7.10 A General Numerical Lesson

The lesson extends beyond the Woodbury identity.

When an expression contains a matrix inverse, one should not automatically conclude that the inverse itself must be explicitly computed. If the inverse appears only through its action on a vector or a small collection of vectors, then solving linear systems is generally the more natural computational interpretation.

Thus, instead of thinking

$$\text{“I need } A^{-1},\text{”}$$

we should often think

$$\boxed{\text{“I need to know how } A^{-1} \text{ acts on these particular vectors.”}}$$

That distinction can turn a theoretically correct formula into a much more efficient numerical procedure.

8 Conclusion

The Sherman–Morrison and Woodbury formulas are more than convenient identities for updating the inverse of a matrix. They illustrate a much broader principle in numerical linear algebra and scientific computing:

When a problem changes, first determine how much of the old computation can still be used.

Many procedures in numerical mathematics, statistics, optimization, and data science eventually lead to a linear system of the form

$$Ax = b.$$

Neither the data nor the mathematical objects constructed from the data remain unchanged forever. New observations may arrive, old observations may be removed, parameters or weights may change, or the underlying model itself may require an update. Consequently, both b and A may change as a computation evolves.

The immediate reaction to such a change should not necessarily be to discard the previous computation and start again. Before starting the machine, one should first examine the mathematical structure of the new problem.

Suppose

$$A \longrightarrow A + \Delta A.$$

Before recomputing everything, we should ask:

What is the structure of ΔA ?

Can it be represented as a rank-one update,

$$\Delta A = uv^T?$$

Can it be represented as a sum of a small number of rank-one updates,

$$\Delta A = u_1v_1^T + \cdots + u_kv_k^T?$$

Or, equivalently, can it be represented as a low-rank update,

$$\Delta A = UV^T, \quad k \ll n?$$

More generally, can the modification be represented in the structured form

$$\Delta A = BCD,$$

where the intermediate dimension m is much smaller than n ?

If the answer to any of these questions is yes, then the Sherman–Morrison or Woodbury identity may allow us to reuse information from the original problem rather than performing the entire computation again.

Thus, the important question is not simply

How do I compute the solution of the new problem?

but rather

How is the new problem related to the problem that I have already solved, and what part of my previous computation can I preserve?

Structure Before Computation

The Sherman–Morrison formula teaches us how to exploit a rank-one change:

$$\boxed{\text{rank-one change} \implies \text{rank-one correction.}}$$

The Woodbury identity extends the same philosophy:

$$\boxed{\text{low-rank change} \implies \text{low-dimensional correction problem.}}$$

For large-scale data science problems, this distinction can be substantial. If a matrix has dimension n but its modification has rank k , where

$$k \ll n,$$

then the Woodbury identity can transfer the essential new computation to a much smaller $k \times k$ system.

The number of entries that have changed is therefore not necessarily the important quantity. A modification may affect a large portion, or even every entry, of a matrix while still possessing low-rank structure. It is the structure and rank of the modification that determine whether these update formulas can be exploited.

Do Not Compute an Inverse Merely Because an Inverse Appears

The numerical interpretation of the Woodbury identity gives us another important lesson.

A mathematical formula containing

$$A^{-1}$$

does not necessarily instruct us to explicitly compute A^{-1} .

If we require

$$A^{-1}b,$$

we may instead solve

$$Ax = b.$$

If we require

$$A^{-1}B,$$

we may instead solve

$$AE = B,$$

or solve one system for each column of B .

Thus, expressions involving an inverse should often be interpreted in terms of the *action* of that inverse:

$$\boxed{A^{-1}r \longleftrightarrow \text{solve } Az = r.}$$

This distinction becomes especially valuable when a factorization of A has already been computed. The same factorization can be reused for multiple right-hand sides rather than constructing the full inverse or factorizing the modified matrix again.

For the general Woodbury modification

$$A + BCD,$$

this reasoning leads to the practical sequence

$$\begin{aligned} Ad &= b, \\ AE &= B, \\ (C^{-1} + DE)y &= Dd, \\ Ax &= b - By. \end{aligned}$$

The large system involving A is handled using computations that can be reused, while the essential effect of the modification is captured by the much smaller system

$$(C^{-1} + DE)y = Dd.$$

Thus, Woodbury is not merely an identity for writing down a new inverse. It provides a way of reorganizing the computation so that the difficult part of the update can be transferred to a lower-dimensional problem.

Think Before Starting the Machine

The broader lesson extends far beyond the Sherman–Morrison and Woodbury formulas.

Modern computers can perform an enormous number of arithmetic operations, but computational power should not replace mathematical reasoning. When a system changes, there is no need to immediately begin another large computation. We should first ask whether the new problem contains information that has already been computed.

Before beginning a computation, one should therefore ask:

What actually changed?

What remained unchanged?

Can the change be represented in a simpler or lower-dimensional form?

Can a previous factorization or solution be reused?

Does the formula really require an inverse, or only the action of that inverse on a few vectors?

Can the large problem be reduced to a much smaller one?

These questions should come *before* the arithmetic begins.

The Sherman–Morrison and Woodbury identities provide a particularly beautiful example of this philosophy. A changing system does not necessarily require an entirely new computation. By identifying the structure of the change, preserving useful previous work, and replacing unnecessary matrix inversions with linear system solves, a seemingly large new problem may become only a small correction to an old one.

The central lesson of this discussion may therefore be summarized as

Before recomputing a problem from the beginning, determine what actually changed and what can still be reused. Exploit the structure of the change, preserve previous computations whenever possible, and reduce the new task before asking the machine to perform it.

In numerical mathematics, the most efficient computation is often not the computation performed fastest by the machine, but the computation that mathematical reasoning allows us to simplify—or avoid altogether.

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